A Multi-Metric Benchmark for VIS–SWIR Image Registration under Cloudy Conditions

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Contributions

- We propose a multi-metric benchmark for VIS–SWIR image registration under cloudy conditions using RGB–SWIR image pairs from the RASMD dataset [1].
- The benchmark compares three alignment approaches: OpenCV, Patch MAGSAC++, and RASMD Aligned.
- Registration quality is evaluated using GOA, EGDA, MIND-SSD, reprojection error, and EPE.

Motivations



Figure 1: VIS (top) and SWIR (bottom) [1]

- This work focuses on **cross-spectral perception** for embedded systems, with image fusion as the final target application.
- By fusing VIS and SWIR images, we aim to combine the robustness of SWIR under adverse conditions with the color and texture information provided by RGB.
- VIS and SWIR images capture the same scene with different spectral responses, producing strong texture and contrast differences.
- However, such fusion is only meaningful if both modalities are accurately **registered**, which makes cross-spectral calibration and geometric alignment a central requirement of the pipeline.

Dataset, Methods and Evaluation

- Experiments were conducted on cloudy RGB–SWIR image pairs from the RASMD dataset [1], including both *aligned* and *unaligned* image pairs.
- The Patch MAGSAC++ pipeline is based on a VIS–SWIR calibration project developed at Institut Pascal [2].
- Sky regions were excluded for all methods because cloudy sky areas are low-texture and spectrally inconsistent between VIS and SWIR.

Table 1: Compared VIS–SWIR registration pipelines.

Step	**OpenCV**	**Patch MAGSAC++** [2]	**RASMD Aligned** [1]
Input pair		*Unaligned VIS–SWIR image pairs*	
Keypoint detection	ORB	SuperPoint	SIFT
Feature matching	BFMatcher	LightGlue	SIFT matching
Outlier rejection	RANSAC	MAGSAC++	RANSAC
Estimated transformation		*Global homography*	

Evaluation Metrics

- GOA and EGDA evaluate multimodal edge consistency.
- MIND-SSD measures multimodal structural similarity [3].
- Reprojection error and EPE evaluate geometric alignment accuracy and were inspired by VIS–IR evaluation strategies proposed in [4].

Table 2: Average registration performance on cloudy VIS–SWIR image pairs.

Method	**GOA** ↑	**EGDA** ↓	**MIND** ↓	**Reproj.** ↓	**EPE** ↓
OpenCV	0.6349	**18.0785**	0.0434	6.4211	9.4048
Patch MAGSAC++	**0.6442**	18.1245	**0.0401**	4.5432	7.1328
RASMD Aligned	0.6426	18.1503	0.0402	**4.3985**	**6.4874**

Qualitative Registration Results

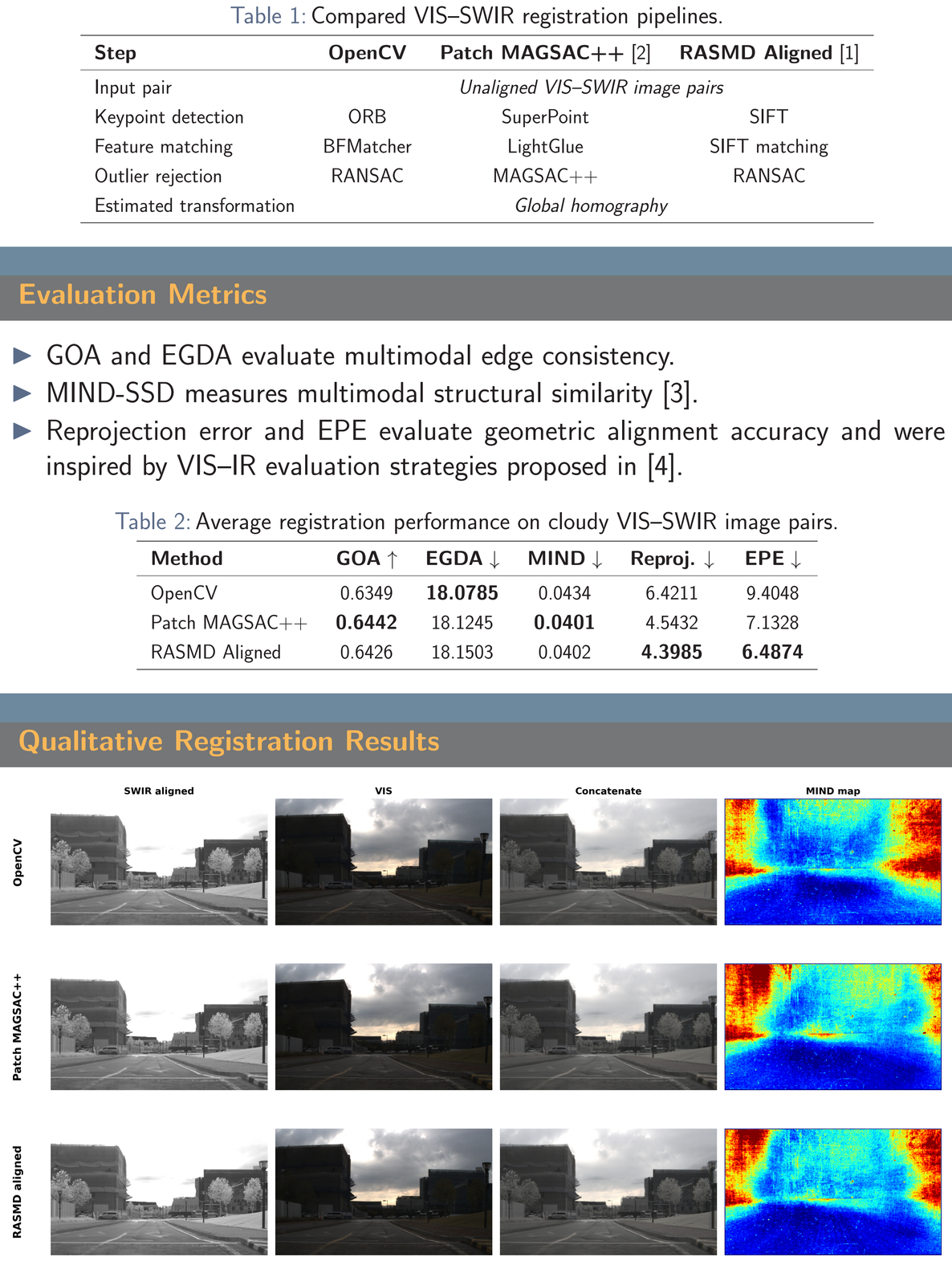


Figure 2: Qualitative comparison of OpenCV, Patch MAGSAC++ and RASMD Aligned on cloudy VIS–SWIR pairs.

Metric Distributions and Frame-wise Analysis

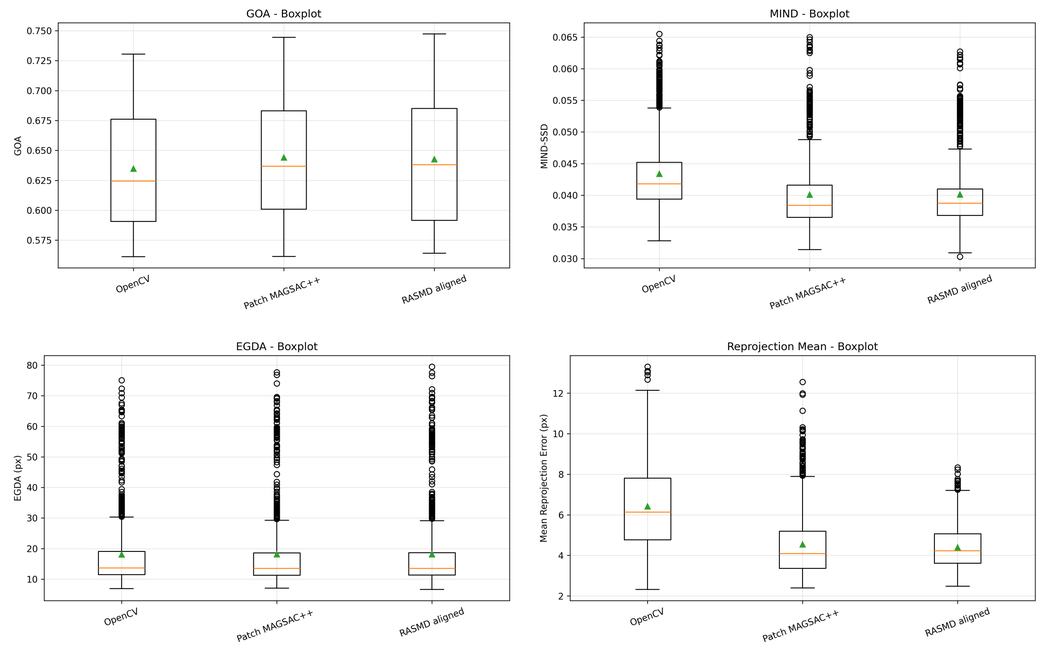


Figure 3: Metric distributions over cloudy VIS–SWIR image pairs.

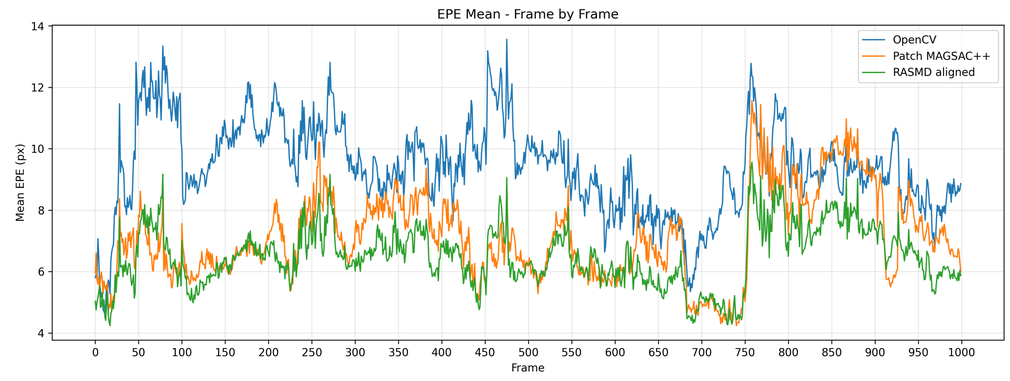


Figure 4: Frame-by-frame EPE evolution over the evaluated sequence.

Conclusion

- The benchmark shows that VIS–SWIR registration must be evaluated using complementary multimodal and geometric metrics.
- Patch MAGSAC++ achieves the best GOA and MIND-SSD scores, while OpenCV obtains the best EGDA performance.
- Patch MAGSAC++ and RASMD Aligned achieve very similar GOA and MIND-SSD values, while RASMD Aligned achieves the lowest reproj. and EPE scores.
- Sky masking improves the reliability of cloudy-scene evaluation and helps localize alignment failures in MIND maps.

Future Work

- Extend the benchmark to sunny, rainy, and snowy VIS–SWIR conditions.
- Include computational efficiency as an additional metric.
- Improve the Patch MAGSAC++ registration pipeline for more robust global homography estimation.
- Compare with other VIS–SWIR calibration approaches, for instance methods based on dense feature extraction.

References

[1] Youngwan Jin et al. "RASMD: RGB And SWIR Multispectral Driving Dataset for Robust Perception in Adverse Conditions". In: *arXiv preprint arXiv:2504.07603* (2025).

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Acknowledgments

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